MILITARY SPECIFICATION

GRENADE, XM68, SAFETY CLIP FOR

1. SCOPE

1.1 This specification covers a safety clip for use on Grenade XM68.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-26	-	Bag, Plastic, Polyethylene, General Purpose

MILITARY

MIL-B-2427	-	Box, Ammunition, Packing, Wood, Nailed
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-B-3106	-	Board Composition, Water Resistant, Solid, for Filler or Cushioning Pads
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

FSC: 1330

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9235366	-	Clip, Safety

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET-9235366	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials.-The materials shall be in accordance with the applicable drawings and specifications.

3.2 Clip, safety.-The safety clip shall comply with all requirements specified on Drawing (Dwg.) 9235366 and with all requirements specified in applicable specifications.

3.3 Release resistance.-The safety clip shall not be disengaged from its assembled position by a load of twenty-five (25) ounces, but shall be completely disengaged from the reassembled position over the lever by a load of eighty (80) ounces, when tested as specified in 4.4.1.

3.4 Workmanship.-The parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified on the contract.

3.6 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the Quality Assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by code number, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Data submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Fifty (50) safety clips
b. Four (4) safety clips or four (4) separate specimens for salt spray.

All parts submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component, fails to comply with any or all of the specified requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article sample inspection. Until a first article sample is accepted, the contractor is, in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "Lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monghly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbers, as required, shall be in accordance with MIL-STD-1168.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used, if approved by the procuring activity. Also at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect, except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Clip, safety (See Dwg. 9235366).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Profile incorrect	Gage	01001
102.	Location of formed loop	Gage	01002
103.	Diameter of wire	Gage	01003
104	Width across straight portions	Gage	01004
Minor	AQL 2.50 Percent		
201.	Width of formed rectangle	Gage	01005
202.	Profile of formed loop	Gage	01006
203.	Inside diameter of curved portion	Gage	01007
204.	Protective finish missing or contains bare spot	Visual	01008
205.	Evidence of poor workmanship (See 3.4)	Visual	01009

4.3.3 Testing.-

4.3.3.1 Release resistance (See 3.3) - Major defect - Code No.02001.- The sampling plan for this test shall be in accordance with Table B-1 Code letter O of MIL-STD-414, using an AQL for the non-disengagement (lower limit) load test of 0.065 percent and an AQL of 1.00 percent for the disengagement (upper limit) load test. The lot shall be rejected if the acceptance criteria of the sampling plan are not met or if any safety clip is disengaged by a load of less than twenty-five (25) ounces. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

4.3.3.2 Salt spray(See Dwg.9235366), Major defect, Code No. 03001). Sampling and rejection shall be in accordance with QQ-P-416. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. It is further required that, although no approval is necessary, the contractor shall notify Picatinny Arsenal, ATTN: ND13, of all instances wherein non-mandatory Army designs are specified and used as such by the contractor as defined in 6.3.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.-

4.4.1 Release resistance.-The safety clip shall be assembled on the inert grenade and fuze assembly specified in 4.3.4. The required load that the clip must retain without dislodgement shall slowly and steadily be applied parallel with the safety pin, in the clip releasing direction, until the clip moves to a position that will allow the lever to function when the safety pin is removed. The applied load that caused disengagement shall be recorded for determination of lot disposition. This test is a non-destructive test. Parts so tested may be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

4.4.2 Salt spray.-The safety clips shall be tested as specified in FED-STD-151, Method 811. (5 percent salt spray solution). This test is a destructive test. Parts so tested shall not be returned to the lot. This test shall be conducted for 96 hours.

5. PREPARATION FOR DELIVERY

5.1 The safety clips shall be packaged, packed and marked in accordance with MIL-STD-1169 and the following additional instructions.

5.1.1 Packaging.-

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging.-Five (5) clips shall be placed in a polyethylene bag conforming to Type II, Grade A, Style 1 of PPP-B-26. Film thickness shall be a minimum of .004 inch. Bag closure shall be by heat or impulse sealing.

5.1.1.1.2 Intermediate packaging.-Five hundred (500) unit packages shall be placed in a fiber box and liner as described for the alternate package in paragraph 3.4.1 of MIL-STD-1169.

5.1.1.2 Level C - Unit packaging.-One hundred (100) clips shall be nested on a paperboard tube with ends capped or taped to prevent clips from sliding off tube. Tubes shall be constructed of kraft or commercial chipboard spiral or convolute wound having an outside diameter of 3/4 inch and a wall thickness of .015 inch minimum. The length of the tube shall be approximately ten (10) inches.

5.1.2 Packing.-

5.1.2.1 Level A.-Shipping containers shall conform to MIL-B-2427, Type 2, Class 2, Grade A. All interior surfaces of the box shall be lined with 1/16 inch thick composition board conforming to MIL-B-3106, Type 1 or 11. Additional board shall be added as required to provide a tight pack.

5.1.2.2 Level C.-Each layer of unit packages shall be separated by a plain fiberboard pad equal in size to the inside length and width of the shipping container less 1/16 inch. Additional pads shall be placed on top of the contents as required to provide a tight pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

5.2 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA-1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections (See 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings, which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors, normally are not required to furnish drawings of commercial inspection equipment. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.- Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1330-A117

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:09Z
Check the source to verify that this is the current version before use.